

EQUIPMENT INSPECTION REPORT - CDU UNIT 1

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Equipment ID: MRPL-PUMP-101-B

Equipment Description: Crude Feed Centrifugal Pump

Operating Unit: CDU-1 (Atmospheric Distillation Unit)

Inspection Date: 2026-09-02

Lead Mechanical Inspector: S. Ramanathan (Emp #44102)

CRITICAL OBSERVATIONS:

- Drive-End (DE) Bearing Housing Temperature: 82 deg C (High Alarm limit: 75 deg C).
- Non-Drive End (NDE) Bearing Temperature: 64 deg C (Normal).
- **Mechanical Seal Zone: Active seal flush oil leakage observed at 15 drops/min.**
- Overall Vibration Velocity: 4.8 mm/s RMS (ISO 10816 Class II Limit: 4.5 mm/s RMS).
- **Acoustic Inspection: Audible high-frequency bearing cage clicking noise.**
- **Casing Integrity: Surface oxidation and oil staining on foundation skirt.**

OPERATIONAL IMPACT & RISK RATING: HIGH

Risk of unexpected bearing seizure and crude feed loss causing CDU furnace trip.

INSPECTOR RECOMMENDATION:

1. Schedule urgent maintenance intervention within 48 hours.
2. Replace Drive-End angular contact ball bearing assembly (SKF 7314 BECBM).
3. Replace mechanical seal cartridge assembly (Plan 53A dual pressurized seal).
4. Perform laser coupling alignment and lube oil flushing before restart.